

Our First Class – Sphere:

heap-puzzle3.cpp

```
5 int *x = new int;
6 int &y = *x;
7
8 y = 4;
9
10 cout << &x << endl;
11 cout << x << endl;
12 cout << *x << endl;
13
14 cout << &y << endl;
15 cout << y << endl;
16 cout << *y << endl;
```

x	Type:	y	Type:
&x	Value:	&y	Value:
x	Value:	y	Value:
*x	Value:	*y	Value:

joinSpheres-byValue.cpp

```
11 /*
12  * Creates a new sphere that contains the exact volume
13  * of the sum of volume of two input spheres.
14  */
15 Sphere joinSpheres(Sphere s1, Sphere s2) {
16     double totalVolume = s1.getVolume() + s2.getVolume();
17
18     double newRadius = std::pow(
19         (3.0 * totalVolume) / (4.0 * 3.141592654),
20         1.0/3.0
21     );
22
23     Sphere result(newRadius);
24
25     return result;
26 }
```

joinSpheres-byPointer.cpp

```
15 Sphere joinSpheres(Sphere *s1, Sphere *s2) {
16     double totalVolume = s1->getVolume() + s2->getVolume();
17
18     double newRadius = std::pow(
19         (3.0 * totalVolume) / (4.0 * 3.141592654),
20         1.0/3.0
21     );
22
23     Sphere result(newRadius);
24
25     return result;
26 }
```

joinSpheres-byReference.cpp

```
15 Sphere joinSpheres(Sphere &s1, Sphere &s2) {
16     double totalVolume = s1.getVolume() + s2.getVolume();
17
18     double newRadius = std::pow(
19         (3.0 * totalVolume) / (4.0 * 3.141592654),
20         1.0/3.0
21     );
22
23     Sphere result(newRadius);
24
25     return result;
26 }
```

	By Value	By Pointer	By Reference
Exactly what is copied when the function is invoked?			
Does modification of the passed in object modify the caller's object?			
Is there always a valid object passed in to the function?			
Speed			
Safety			

Using the const keyword

1. [Function Parameters]:

joinSpheres-byValue-const.cpp	
15	Sphere joinSpheres(const Sphere s1, const Sphere s2)

joinSpheres-byPointer-const.cpp	
15	Sphere joinSpheres(Sphere const *s1, Sphere const *s2)

joinSpheres-byReference-const.cpp	
15	Sphere joinSpheres(const Sphere &s1, const Sphere &s2)

Using the const keyword

2. [Classes]:

sphere.h		sphere.cpp	
1	#ifndef SPHERE_H	1	/* ... */
2	#define SPHERE_H		
3			
4	class Sphere {		
5	public:		
6	Sphere();		
7	Sphere(double r);		
8			
9	double getRadius();		
10	double getVolume();		
11			
12	void setRadius(double r);		
13			
14	private:		
15	double r_;		
16	};		
17			
18	#endif		

Big Idea: Copy Constructor

sphere.h		sphere.cpp	
1	#ifndef SPHERE_H		/* ... */
2	#define SPHERE_H		
3			
4	class Sphere {		
5	public:		
6	Sphere();		
7	Sphere(double r);		
8			
9			
10			
11			
12			
...	/* getRadius, getVolume,		
	setRadius, and r_ */		/* ... */

Bringing Concepts Together:

How many times do our different joinSphere files call each constructor?

	By Value	By Pointer	By Reference
Sphere()			
Sphere(double)			
Sphere(const Sphere &)			

CS 225 – Things To Be Doing:

1. Quiz 1 is coming up this Thursday, see BB for details
2. lab_intro due Sunday
3. mp1 due Monday after the winter break